

Chapter 2

Group Theory Basics

Groups are the central object of abstract algebra, capturing the essence of symmetry and invertible operations. The definition is deceptively simple—three axioms—yet groups appear everywhere: in geometry (symmetries), number theory (modular arithmetic), topology (fundamental groups), physics (gauge theories), and even chemistry (molecular symmetries). This lecture develops the foundations rigorously.

2.1 The group axioms

Definition 2.1.1: Groups

A **group** is a set G together with a map

$$m : G \times G \rightarrow G$$

called a **law of composition** (also **binary operation**). Writing ab for $m(a, b)$, the law of composition must satisfy three axioms:

1. **Associativity:** $\forall a, b, c \in G, (ab)c = a(bc)$.
2. **Identity:** $\exists e \in G$ such that $ea = ae = a$ for all $a \in G$.
3. **Inverses:** $\forall a \in G, \exists b \in G$ such that $ab = ba = e$.

We write (G, m) or simply G for the group, and call $|G|$ the **order** of G .

The axioms have immediate but important consequences that we now prove carefully.

Proposition 2.1.1 Uniqueness of identity

The identity element e is unique.

Proof: Uniqueness of identity

Suppose e and e' are both identities. Then $e = ee'$ (since e' is an identity) and $ee' = e'$ (since e is an identity). Thus $e = e'$. $\square$

Proposition 2.1.2 Uniqueness of inverses

For each $a \in G$, the inverse is unique. We denote it a^{-1} (or $-a$ in additive notation).

Proof: Uniqueness of inverses

Suppose b and c are both inverses of a , so $ab = ba = e$ and $ac = ca = e$. Then:

$$b = be = b(ac) = (ba)c = ec = c.$$

□

Proposition 2.1.3 Cancellation laws

In a group G :

1. **Left cancellation:** $ab = ac \implies b = c$.
2. **Right cancellation:** $ba = ca \implies b = c$.

Proof: Proof of cancellation

For left cancellation: if $ab = ac$, multiply both sides on the left by a^{-1} : $a^{-1}(ab) = a^{-1}(ac)$, so $(a^{-1}a)b = (a^{-1}a)c$, giving $eb = ec$, i.e., $b = c$. Right cancellation is similar. □

Proposition 2.1.4 Inverse of products

For all $a, b \in G$: $(ab)^{-1} = b^{-1}a^{-1}$. Note the reversal of order!

Proof: Proof of inverse formula

We verify $(ab)(b^{-1}a^{-1}) = e$: $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e$. Similarly for the other side. □

Proposition 2.1.5 Inverse of inverse

For all $a \in G$: $(a^{-1})^{-1} = a$.

Proof: Proof of double inverse

By definition, a^{-1} satisfies $a \cdot a^{-1} = e$. This says a is an inverse of a^{-1} , so by uniqueness, $(a^{-1})^{-1} = a$. □

Note:-

It is customary to use additive notation $(+, 0, -a)$ for abelian groups and multiplicative notation $(\cdot, 1$ or $e, a^{-1})$ in general. This does not mean that multiplication implies non-commutativity!

2.1.1 Group variants and related structures

Groups sit in a hierarchy of algebraic structures with fewer or more axioms.

Definition 2.1.2: Abelian groups

A group G is **abelian** (or **commutative**) if $ab = ba$ for all $a, b \in G$.

Definition 2.1.3: Monoid

A **monoid** is a set with an associative binary operation and an identity element, but inverses are not required. Example: $(\mathbb{N}, +)$ with identity 0, or $(\mathbb{Z}, \times)$ with identity 1.

Definition 2.1.4: Semigroup

A **semigroup** is a set with an associative binary operation—no identity or inverses required. Example: $(\mathbb{Z}_{>0}, +)$.

Note:-

There's a pattern here: Group = Monoid + Inverses = Semigroup + Identity + Inverses. Category theory provides a unifying framework: a group is a category with one object where every morphism is invertible; a monoid is a category with one object (morphisms need not be invertible).

2.2 Fundamental examples

2.2.1 Integers and modular arithmetic

Example 2.2.1 (The integers)

The set $(\mathbb{Z}, +)$ is an abelian group with identity 0 and inverse $-n$ for each n . This is the prototypical infinite cyclic group.

Example 2.2.2 (Integers mod n)

$\mathbb{Z}/n\mathbb{Z} = \{0, 1, 2, \dots, n-1\}$ with addition mod n :

$$a + b := \begin{cases} a + b & \text{if } a + b < n \\ a + b - n & \text{if } a + b \geq n. \end{cases}$$

Identity is 0, inverse of a is $n - a \pmod{n}$. This is an abelian group of order n .

Proposition 2.2.1 Units mod n

The set $(\mathbb{Z}/n\mathbb{Z})^\times = \{a \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}$ forms a group under multiplication mod n . Its order is $\varphi(n)$, Euler's totient function.

Proof: Proof that units form a group

Closure: If $\gcd(a, n) = \gcd(b, n) = 1$, then $\gcd(ab, n) = 1$ (if a prime p divides both ab and n , it divides a or b by Euclid's lemma—which we cite ahead of its proof below—contradicting coprimality).

Identity: 1 is coprime to n .

Inverses: If $\gcd(a, n) = 1$, then by Bézout's identity (cited here ahead of its proof in the subsection below), $\exists r, s$ with $ra + sn = 1$. Thus $ra \equiv 1 \pmod{n}$, so r is a multiplicative inverse. $\square$

2.2.2 Multiplicative groups of fields

Example 2.2.3 (Multiplicative groups)

$\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, and $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ are abelian groups under multiplication, with identity 1 and inverse $1/x$.

Example 2.2.4 (The circle group)

The unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is a group under multiplication. It is isomorphic to $\mathbb{R}/\mathbb{Z}$ via $t \mapsto e^{2\pi i t}$. This is a compact abelian group and plays a role in Fourier analysis and representation theory.

Example 2.2.5 (Roots of unity)

For each $n \geq 1$, the n th roots of unity $\mu_n = \{z \in \mathbb{C} : z^n = 1\} = \{e^{2\pi i k/n} : k = 0, 1, \dots, n-1\}$ form a cyclic group of order n under multiplication. We have $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$.

2.2.3 Symmetric and permutation groups

Definition 2.2.1: Permutation group

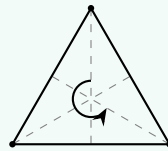
A **permutation** of a set X is a bijection $\sigma : X \rightarrow X$. The set of all permutations of X forms a group $\text{Sym}(X)$ under composition, with identity id_X and inverse σ^{-1} .

Definition 2.2.2: Symmetric group

The **symmetric group** on n letters is $S_n = \text{Sym}(\{1, 2, \dots, n\})$. It has order $n!$ and is non-abelian for $n \geq 3$.

Example 2.2.6 (Symmetries of a triangle)

S_3 can be visualized as symmetries of an equilateral triangle: 3 rotations (including identity) and 3 reflections.



Labeling vertices 1, 2, 3, rotations give permutations $e, (123), (132)$ and reflections give $(12), (13), (23)$. Thus $|S_3| = 6$.

2.2.4 Matrix groups

Definition 2.2.3: General linear group

The **general linear group** $GL_n(\mathbb{F})$ is the group of invertible $n \times n$ matrices over a field $\mathbb{F}$, under matrix multiplication. A matrix is invertible iff its determinant is nonzero.

Definition 2.2.4: Special linear group

The **special linear group** $SL_n(\mathbb{F}) = \{A \in GL_n(\mathbb{F}) : \det(A) = 1\}$ is a subgroup of $GL_n(\mathbb{F})$.

Note:-

Matrix groups are our first examples of **Lie groups**—groups that are also smooth manifolds. They are, in my opinion, the most exciting part of representation theory: a representation of G is a homomorphism $\rho : G \rightarrow GL_n(\mathbb{F})$, making G act on $\mathbb{F}^n$ by linear transformations.

Example 2.2.7 (Non-abelian example)

In $GL_2(\mathbb{R})$, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Then:

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

So $GL_2(\mathbb{R})$ is non-abelian.

2.3 Constructions: Products and sums

How do we build new groups from existing ones?

2.3.1 Direct products

Definition 2.3.1: Direct product

Given groups G and H , the **direct product** $G \times H$ is the Cartesian product with componentwise operation:

$$(a, b) \cdot (a', b') = (aa', bb').$$

Identity is (e_G, e_H) , inverse of (a, b) is (a^{-1}, b^{-1}) .

Proposition 2.3.1 Direct product is a group

The direct product $G \times H$ satisfies all group axioms. Moreover, $|G \times H| = |G| \cdot |H|$ (for finite groups).

Proof: Verification of group axioms for $G \times H$

Associativity: $((a, b)(a', b'))(a'', b'') = (aa', bb')(a'', b'') = (aa'a'', bb'b'')$. Similarly, $(a, b)((a', b')(a'', b'')) = (a, b)(a'a'', b'b'') = (aa'a'', bb'b'')$. Equal by associativity in G and H .

Identity: $(e_G, e_H)(a, b) = (e_G a, e_H b) = (a, b)$, and similarly on the right.

Inverses: $(a, b)(a^{-1}, b^{-1}) = (aa^{-1}, bb^{-1}) = (e_G, e_H)$. □

Example 2.3.1

We have $\mathbb{Z}/2 \times \mathbb{Z}/3 \cong \mathbb{Z}/6$ (since $\gcd(2, 3) = 1$), but $\mathbb{Z}/2 \times \mathbb{Z}/2 \not\cong \mathbb{Z}/4$ (the former has no element of order 4).

This generalizes to arbitrary (even infinite) collections of groups.

Definition 2.3.2: Arbitrary direct products

For a family of groups $\{G_i\}_{i \in I}$, the **direct product** is:

$$\prod_{i \in I} G_i = \{(a_i)_{i \in I} : a_i \in G_i\}$$

with componentwise operations.

Definition 2.3.3: Direct sum

The **direct sum** (for abelian groups) is the subgroup:

$$\bigoplus_{i \in I} G_i = \{(a_i)_{i \in I} \in \prod_{i \in I} G_i : a_i = e_{G_i} \text{ for all but finitely many } i\}.$$

Note:-

For finite index sets, $\prod = \bigoplus$. For infinite sets, they differ crucially. "O-plus" notation will become important when we move onto linear algebra.

Example 2.3.2 (Power series vs polynomials)

Taking $G_i = (\mathbb{R}, +)$ for all $i \in \mathbb{N}$:

$$\prod_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[[x]] \quad (\text{formal power series under addition})$$

$$\bigoplus_{i=0}^{\infty} \mathbb{R} \cong \mathbb{R}[x] \quad (\text{polynomials under addition})$$

A power series $\sum_{i=0}^{\infty} a_i x^i$ can have infinitely many nonzero terms; a polynomial cannot.

Question 1

1. Prove that $\mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/mn$ if and only if $\gcd(m, n) = 1$.
2. Find all groups of order 4 up to isomorphism.
3. Prove that $G \times H \cong H \times G$ for any groups G, H .

Solution

(1) ($\Leftarrow$) Assume $\gcd(m, n) = 1$. The element $(1, 1) \in \mathbb{Z}/m \times \mathbb{Z}/n$ has order $\text{lcm}(m, n) = mn$ (since the order of (a, b) is $\text{lcm}(\text{ord}(a), \text{ord}(b))$). Thus $\mathbb{Z}/m \times \mathbb{Z}/n$ contains an element of order $mn = |G|$, so it's cyclic, hence $\cong \mathbb{Z}/mn$.

($\Rightarrow$) If $d = \gcd(m, n) > 1$, then for any $(a, b) \in \mathbb{Z}/m \times \mathbb{Z}/n$, we have $\frac{mn}{d} \cdot (a, b) = (\frac{mn}{d}a, \frac{mn}{d}b) = (0, 0)$ since $m \mid \frac{mn}{d}$ and $n \mid \frac{mn}{d}$. So every element has order dividing $\frac{mn}{d} < mn$, meaning no element has order mn . But $\mathbb{Z}/mn$ has an element of order mn , so they're not isomorphic.

(2) Let G have order 4. Every element has order dividing 4, so orders are 1, 2, or 4.

Case 1: G has an element a of order 4. Then $G = \{e, a, a^2, a^3\} = \langle a \rangle \cong \mathbb{Z}/4$.

Case 2: Every non-identity element has order 2. Then $G = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$. What is ab ? It's not e (else $a = b^{-1} = b$), not a (else $b = e$), not b (else $a = e$). So $ab = c$. Similarly $ba = c$ (check: $(ab)(ba) = a(bb)a = a^2 = e$, so $ba = (ab)^{-1} = ab$ since ab has order 2). So G is abelian, and $G \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four-group).

(3) Define $\varphi : G \times H \rightarrow H \times G$ by $\varphi(g, h) = (h, g)$. This is clearly a bijection. It's a homomorphism: $\varphi((g_1, h_1)(g_2, h_2)) = \varphi(g_1 g_2, h_1 h_2) = (h_1 h_2, g_1 g_2) = (h_1, g_1)(h_2, g_2) = \varphi(g_1, h_1)\varphi(g_2, h_2)$.

2.4 Subgroups

Definition 2.4.1: Subgroup

A **subgroup** of a group G is a nonempty subset $H \subseteq G$ such that:

1. **Closure:** $a, b \in H \implies ab \in H$;
2. **Inverses:** $a \in H \implies a^{-1} \in H$.

We write $H \leq G$ (or $H < G$ if $H \neq G$, a **proper subgroup**).

Proposition 2.4.1 Subgroup criterion

A nonempty subset $H \subseteq G$ is a subgroup iff $a, b \in H \implies ab^{-1} \in H$.

Proof: One-step subgroup test

($\Rightarrow$) If $H \leq G$ and $a, b \in H$, then $b^{-1} \in H$ (inverses) and $ab^{-1} \in H$ (closure).

($\Leftarrow$) Assume the condition. Since $H \neq \emptyset$, pick $a \in H$. Then $e = aa^{-1} \in H$. For any $b \in H$, $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we have $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$. $\square$

Example 2.4.1 (Subgroups of $\mathbb{Z}$)

Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\}$ for some $n \geq 0$. These form a chain: $\mathbb{Z} = 1\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset 8\mathbb{Z} \supset \dots$ (not all subgroups are comparable, though: $2\mathbb{Z}$ and $3\mathbb{Z}$ are incomparable).

Theorem 2.4.1 Subgroups of $\mathbb{Z}$

Let H be a subgroup of $(\mathbb{Z}, +)$. Then either $H = \{0\}$ or $H = n\mathbb{Z}$ where n is the smallest positive element of H .

Proof: Classification of subgroups of $\mathbb{Z}$

If $H = \{0\}$, we're done. Otherwise, H contains some nonzero element a ; then $-a \in H$ too, so H contains a positive element. Let n be the smallest positive element of H .

We show $H = n\mathbb{Z}$. Clearly $n\mathbb{Z} \subseteq H$ since H is closed under addition: $n \in H \implies 2n = n + n \in H \implies 3n \in H \implies \dots$, and $-n \in H \implies -2n \in H \implies \dots$

For $H \subseteq n\mathbb{Z}$: take any $m \in H$. By division algorithm, $m = qn + r$ with $0 \leq r < n$. Then $r = m - qn \in H$ (since $m, qn \in H$). By minimality of n , we must have $r = 0$, so $m = qn \in n\mathbb{Z}$. $\square$

Proposition 2.4.2 Greatest common divisor

Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then:

1. $\mathbb{Z}a + \mathbb{Z}b = \{ra + sb : r, s \in \mathbb{Z}\} = d\mathbb{Z}$;
2. d divides both a and b ;
3. If n divides both a and b , then n divides d ;
4. $\exists r, s \in \mathbb{Z}$ such that $d = ra + sb$ (Bézout's identity).

Proof: Proof of GCD properties

The set $S = \mathbb{Z}a + \mathbb{Z}b$ is a subgroup of $\mathbb{Z}$ (verify: closed under addition and negation). By the theorem, $S = d\mathbb{Z}$ for some $d \geq 0$. Since $a = 1 \cdot a + 0 \cdot b \in S$ and $b \in S$, and $S = d\mathbb{Z}$, we have $d \mid a$ and $d \mid b$. Since $d \in S$, we have $d = ra + sb$ for some r, s . If $n \mid a$ and $n \mid b$, then $n \mid (ra + sb) = d$. These properties characterize $d = \gcd(a, b)$. $\square$

Corollary 2.4.1 Coprimality criterion

a and b are coprime (i.e., $\gcd(a, b) = 1$) iff $\exists r, s \in \mathbb{Z}$ with $ra + sb = 1$.

Corollary 2.4.2 Euclid's lemma

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: Proof of Euclid's lemma

Suppose $p \nmid a$. Since p is prime, $\gcd(p, a) = 1$. By Bézout, $\exists r, s$ with $rp + sa = 1$. Multiply by b : $rpb + sab = b$. Since $p \mid rpb$ and $p \mid ab$, we have $p \mid b$. $\square$

2.4.1 Generated subgroups and cyclic groups

Definition 2.4.2: Subgroup generated by a set

For $S \subseteq G$, the **subgroup generated by S** , denoted $\langle S \rangle$, is the smallest subgroup of G containing S . Explicitly:

$$\langle S \rangle = \{s_1^{\epsilon_1} s_2^{\epsilon_2} \cdots s_k^{\epsilon_k} : k \geq 0, s_i \in S, \epsilon_i \in \{1, -1\}\}.$$

Definition 2.4.3: Cyclic groups

A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$. The element a is called a **generator**.

Definition 2.4.4: Order of an element

The **order** of $a \in G$, denoted $\text{ord}(a)$ or $|a|$, is the smallest positive integer n such that $a^n = e$, or ∞ if no such n exists.

Proposition 2.4.3 Structure of cyclic groups

Let $G = \langle a \rangle$.

1. If $\text{ord}(a) = \infty$, then $G \cong \mathbb{Z}$ via $n \mapsto a^n$.
2. If $\text{ord}(a) = n < \infty$, then $G = \{e, a, a^2, \dots, a^{n-1}\}$ and $G \cong \mathbb{Z}/n\mathbb{Z}$.

Proof: Proof of cyclic group structure

Define $\varphi : \mathbb{Z} \rightarrow G$ by $\varphi(k) = a^k$. This is a homomorphism: $\varphi(k+l) = a^{k+l} = a^k a^l = \varphi(k)\varphi(l)$. It's surjective since $G = \langle a \rangle$.

Case 1: $\text{ord}(a) = \infty$. If $\varphi(k) = \varphi(l)$ with $k > l$, then $a^{k-l} = e$, contradicting infinite order. So φ is injective, hence an isomorphism $\mathbb{Z} \xrightarrow{\sim} G$.

Case 2: $\text{ord}(a) = n$. Then $\ker(\varphi) = \{k \in \mathbb{Z} : a^k = e\}$. Since $a^n = e$, we have $n\mathbb{Z} \subseteq \ker(\varphi)$. Conversely, if $a^k = e$, write $k = qn + r$ with $0 \leq r < n$. Then $a^r = a^{k-qn} = a^k (a^n)^{-q} = e$. By minimality of n , $r = 0$, so $k \in n\mathbb{Z}$. Thus $\ker(\varphi) = n\mathbb{Z}$, and $G \cong \mathbb{Z}/n\mathbb{Z}$ by the First Isomorphism Theorem (this will be introduced later but to oversimplify: modding out by the kernel turns a surjective homomorphism into an isomorphism onto its image). $\square$

Proposition 2.4.4 Subgroups of cyclic groups

Every subgroup of a cyclic group is cyclic. If $G = \langle a \rangle$ has order n , the subgroups of G are exactly $\langle a^d \rangle$ for divisors $d \mid n$, and $|\langle a^d \rangle| = n/d$.

Proof: Subgroups of cyclic groups

Let $H \leq \langle a \rangle$. If $H = \{e\}$, it's cyclic. Otherwise, let d be the smallest positive integer with $a^d \in H$. We claim $H = \langle a^d \rangle$.

Clearly $\langle a^d \rangle \subseteq H$. For the reverse: take $a^k \in H$. Write $k = qd + r$ with $0 \leq r < d$. Then $a^r = a^k (a^d)^{-q} \in H$. By minimality of d , $r = 0$, so $a^k = (a^d)^q \in \langle a^d \rangle$.

If $|G| = n$, then a^d has order $n/\gcd(d, n)$. The subgroups correspond to divisors via $d \leftrightarrow \langle a^{n/d} \rangle$ of order d . $\square$

Question 2

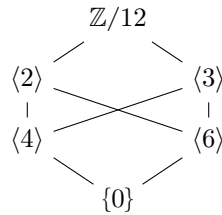
1. Find all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and draw the subgroup lattice.

2. Prove that $\mathbb{Z}/p\mathbb{Z}$ has no proper nontrivial subgroups for p prime.
3. How many generators does $\mathbb{Z}/n\mathbb{Z}$ have?
4. Prove: if G is a group where every element satisfies $x^2 = e$, then G is abelian.

Solution

(1) Divisors of 12: 1, 2, 3, 4, 6, 12. Subgroups: $\langle 0 \rangle = \{0\}$, $\langle 6 \rangle = \{0, 6\}$, $\langle 4 \rangle = \{0, 4, 8\}$, $\langle 3 \rangle = \{0, 3, 6, 9\}$, $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$, $\langle 1 \rangle = \mathbb{Z}/12$.

Lattice (larger subgroups higher):



(2) By Lagrange's theorem (cited here ahead of its proof in the next lecture), the order of any subgroup divides p . So subgroups have order 1 or p , meaning $\{0\}$ or $\mathbb{Z}/p$.

(3) a generates $\mathbb{Z}/n$ iff $\text{ord}(a) = n$ iff $\gcd(a, n) = 1$. The count is $\varphi(n)$, Euler's totient.

(4) For any $a, b \in G$: $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$ (using $x^{-1} = x$ since $x^2 = e$). So G is abelian.

2.5 Homomorphisms

Definition 2.5.1: Homomorphism

A **homomorphism** $\varphi : G \rightarrow H$ between groups is a map satisfying

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

Proposition 2.5.1 Basic properties of homomorphisms

Let $\varphi : G \rightarrow H$ be a homomorphism.

1. $\varphi(e_G) = e_H$.
2. $\varphi(a^{-1}) = \varphi(a)^{-1}$ for all $a \in G$.
3. $\varphi(a^n) = \varphi(a)^n$ for all $n \in \mathbb{Z}$.
4. If $K \leq G$, then $\varphi(K) \leq H$.
5. If $L \leq H$, then $\varphi^{-1}(L) \leq G$.

Proof: Proofs of homomorphism properties

(1) $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)\varphi(e_G)$. Multiply both sides by $\varphi(e_G)^{-1}$: $e_H = \varphi(e_G)$.

(2) $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$. So $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

(3) For $n > 0$: induction. For $n = 0$: part (1). For $n < 0$: $\varphi(a^n) = \varphi((a^{-1})^{-n}) = \varphi(a^{-1})^{-n} = (\varphi(a)^{-1})^{-n} = \varphi(a)^n$.

(4) For $\varphi(a), \varphi(b) \in \varphi(K)$ (with $a, b \in K$): $\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(K)$ since $ab^{-1} \in K$.

(5) For $a, b \in \varphi^{-1}(L)$: $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in L$ (since $L \leq H$), so $ab^{-1} \in \varphi^{-1}(L)$. $\square$

Note:-

The homomorphism condition can be expressed as a commutative diagram: the two paths $G \times G \xrightarrow{m_G} G \xrightarrow{\varphi} H$ and $G \times G \xrightarrow{\varphi \times \varphi} H \times H \xrightarrow{m_H} H$ give the same result.

$$\begin{array}{ccc} G \times G & \xrightarrow{\varphi \times \varphi} & H \times H \\ m_G \downarrow & & \downarrow m_H \\ G & \xrightarrow{\varphi} & H \end{array}$$

Definition 2.5.2: Types of homomorphisms

Let $\varphi : G \rightarrow H$ be a homomorphism.

1. φ is a **monomorphism** (or **embedding**) if it's injective.
2. φ is an **epimorphism** if it's surjective.
3. φ is an **isomorphism** if it's bijective. We write $G \cong H$.
4. An isomorphism $G \rightarrow G$ is called an **automorphism**. The set of all automorphisms of G , denoted $\text{Aut}(G)$, forms a group under composition.

Example 2.5.1 (Important(ish) homomorphisms)

1. **Reduction mod n** : $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $\pi(a) = a \bmod n$. Surjective, kernel $n\mathbb{Z}$.
2. **Quotient mod n** : If $n \mid m$, then $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $a \mapsto a \bmod n$. Example: last-two-digits to last-digit, $\mathbb{Z}/100 \rightarrow \mathbb{Z}/10$.
3. **Determinant**: $\det : \text{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}^*$. Homomorphism since $\det(AB) = \det(A)\det(B)$. Kernel is $\text{SL}_n(\mathbb{R})$.
4. **Exponential**: $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \times)$, $x \mapsto e^x$. Isomorphism with inverse $\ln$.
5. **Complex exponential**: $\exp : (\mathbb{R}, +) \rightarrow (S^1, \times)$, $t \mapsto e^{2\pi it}$. Surjective, kernel $\mathbb{Z}$. Induces $\mathbb{R}/\mathbb{Z} \cong S^1$.

2.5.1 Isomorphisms and invariants

Definition 2.5.3: Isomorphic groups

Groups G and H are **isomorphic**, written $G \cong H$, if there exists an isomorphism $\varphi : G \rightarrow H$. Isomorphic groups are “the same” for all group-theoretic purposes.

What properties are preserved by isomorphisms? These are called invariants.

Proposition 2.5.2 Isomorphism invariants

If $G \cong H$, then:

1. $|G| = |H|$ (same order);
2. G is abelian iff H is abelian;
3. G is cyclic iff H is cyclic;

4. For each n , G and H have the same number of elements of order n ;
5. For each n , G and H have the same number of subgroups of order n .

Proof: Why these are invariants

Let $\varphi : G \xrightarrow{\sim} H$ be an isomorphism.

(1) Bijections preserve cardinality.

(2) If G is abelian and $h_1, h_2 \in H$, write $h_i = \varphi(g_i)$. Then $h_1 h_2 = \varphi(g_1) \varphi(g_2) = \varphi(g_1 g_2) = \varphi(g_2 g_1) = \varphi(g_2) \varphi(g_1) = h_2 h_1$.

(3) If $G = \langle a \rangle$, then $H = \varphi(G) = \varphi(\langle a \rangle) = \langle \varphi(a) \rangle$.

(4) $\text{ord}(\varphi(a)) = \text{ord}(a)$ since $\varphi(a)^n = \varphi(a^n) = e_H \iff a^n = e_G$.

(5) φ induces a bijection between subgroups of G and subgroups of H . □

Example 2.5.2 (Non-isomorphic groups of same order)

$\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ both have order 4, but:

- $\mathbb{Z}/4$ has an element of order 4 (namely, 1);
- $\mathbb{Z}/2 \times \mathbb{Z}/2$ has no element of order 4 (all non-identity elements have order 2).

So they're not isomorphic. These are the only groups of order 4 up to isomorphism.

Example 2.5.3 (All groups of order 2 are isomorphic)

If $|G| = 2$, write $G = \{e, x\}$. Then $x \neq e$, so $x^2 \neq x$ (else $x = e$). Thus $x^2 = e$, and $G = \langle x \rangle \cong \mathbb{Z}/2$.

Question 3

1. Prove that $\text{Aut}(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.
2. Find $\text{Aut}(\mathbb{Z})$.
3. Prove that if $\varphi : G \rightarrow H$ is an isomorphism, then $\varphi^{-1} : H \rightarrow G$ is also an isomorphism.
4. Prove that “is isomorphic to” is an equivalence relation on groups.

Solution

(1) An automorphism φ of $\mathbb{Z}/n$ is determined by $\varphi(1)$, since $\varphi(k) = k\varphi(1)$. We need $\varphi(1)$ to generate $\mathbb{Z}/n$, i.e., $\gcd(\varphi(1), n) = 1$, i.e., $\varphi(1) \in (\mathbb{Z}/n)^\times$. Conversely, any $a \in (\mathbb{Z}/n)^\times$ defines an automorphism $\varphi_a : k \mapsto ka$. The map $a \mapsto \varphi_a$ is an isomorphism $(\mathbb{Z}/n)^\times \rightarrow \text{Aut}(\mathbb{Z}/n)$ (check it's a homomorphism and bijection).

(2) An automorphism of $\mathbb{Z}$ is determined by $\varphi(1)$. We need $\langle \varphi(1) \rangle = \mathbb{Z}$, so $\varphi(1) = \pm 1$. Thus $\text{Aut}(\mathbb{Z}) = \{\text{id}, -\text{id}\} \cong \mathbb{Z}/2$.

(3) φ^{-1} is a bijection. For $h_1, h_2 \in H$, let $g_i = \varphi^{-1}(h_i)$. Then $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1) \varphi(g_2)) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1) \varphi^{-1}(h_2)$.

(4) Reflexive: $\text{id}_G : G \rightarrow G$ is an isomorphism. Symmetric: if $\varphi : G \rightarrow H$ is an isomorphism, so is $\varphi^{-1} : H \rightarrow G$ by part (3). Transitive: if $\varphi : G \rightarrow H$ and $\psi : H \rightarrow K$ are isomorphisms, so is $\psi \circ \varphi : G \rightarrow K$.

2.6 The order of an element and its properties

We've introduced order; now we develop its theory more fully.

Proposition 2.6.1 Powers and order

Let $a \in G$ have finite order n . Then:

1. $a^k = e \iff n \mid k$.
2. $a^i = a^j \iff n \mid (i - j)$.
3. The elements $e, a, a^2, \dots, a^{n-1}$ are all distinct.

Proof: Properties of order

(1) ($\Leftarrow$) If $k = mn$, then $a^k = (a^n)^m = e^m = e$. ($\Rightarrow$) Write $k = qn + r$ with $0 \leq r < n$. Then $e = a^k = a^{qn+r} = (a^n)^q a^r = a^r$. By minimality of n , $r = 0$, so $n \mid k$.

(2) $a^i = a^j \iff a^{i-j} = e \iff n \mid (i - j)$ by (1).

(3) If $a^i = a^j$ for $0 \leq i < j < n$, then $a^{j-i} = e$ with $0 < j - i < n$, contradicting minimality of n . $\square$

Proposition 2.6.2 Order of powers

Let $\text{ord}(a) = n$. Then $\text{ord}(a^k) = \frac{n}{\gcd(k, n)}$.

Proof: Order of powers

Let $d = \gcd(k, n)$. We show $\text{ord}(a^k) = n/d$.

First, $(a^k)^{n/d} = a^{kn/d} = (a^n)^{k/d} = e$, so $\text{ord}(a^k) \mid n/d$.

Conversely, let $m = \text{ord}(a^k)$. Then $a^{km} = e$, so $n \mid km$. Write $k = d \cdot k'$ and $n = d \cdot n'$ where $\gcd(k', n') = 1$. Then $dn' \mid dk'm$, so $n' \mid k'm$. Since $\gcd(k', n') = 1$, we have $n' \mid m$, i.e., $n/d \mid m$. Thus $m = n/d$. $\square$

Corollary 2.6.1 Generators of cyclic groups

In $\langle a \rangle$ of order n , the element a^k is a generator iff $\gcd(k, n) = 1$.

Example 2.6.1

In $\mathbb{Z}/12$, the generators are 1, 5, 7, 11 (the four elements coprime to 12). The element 4 has order $12/\gcd(4, 12) = 12/4 = 3$.

Question 4

1. In a group G , if $a^{12} = e$ and $a^8 = e$, what are the possible orders of a ?
2. Prove: if $\text{ord}(a) = n$ and $\text{ord}(b) = m$ with $\gcd(n, m) = 1$ and $ab = ba$, then $\text{ord}(ab) = nm$.
3. Find the order of $(2, 3)$ in $\mathbb{Z}/6 \times \mathbb{Z}/8$.

Solution

(1) $\text{ord}(a)$ divides both 12 and 8, so $\text{ord}(a) \mid \gcd(12, 8) = 4$. Possible orders: 1, 2, 4.

(2) Let $k = \text{ord}(ab)$. Then $e = (ab)^{nm} = a^{nm}b^{nm} = (a^n)^m(b^m)^n = e$ (using commutativity), so $k \mid nm$.

Now $e = (ab)^{kn} = a^{kn}b^{kn} = b^{kn}$ (since $a^n = e$), so $m \mid kn$. Since $\gcd(m, n) = 1$, $m \mid k$. Similarly, $n \mid k$. Since $\gcd(m, n) = 1$, $mn \mid k$. Thus $k = nm$.

(3) $\text{ord}(2, 3) = \text{lcm}(\text{ord}(2), \text{ord}(3)) = \text{lcm}(6/\gcd(2, 6), 8/\gcd(3, 8)) = \text{lcm}(3, 8) = 24$.